Towards quantisation of Boussinesq solitons: Two-body scattering on the good branch and its Toda validation tests

Project *quantum_boussinesq_solitons* — PI: Gustav Delius

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Abstract

(c) On the good Boussinesq branch the classical scattering phase is known. At one loop the conditional soliton trace gives a sound-subtracted one-body slope $C_g - s_1 = 1/(4\pi)$. The response of the two-soliton phase to a finite elastic counterterm is calculable at fixed momenta. An interior heavy soliton has an on-shell pair of light daughters and a localized classical growing mode. The light branch is consequently the starting point for a scattering sector whose quantum closure remains open **(d)**.

(c) Three-body clustering, defect consistency and the one-body shift charge all leave the coefficient of $P_1M_2 - P_2M_1$ undetermined. A classical current deformation changes the two-body action in precisely this direction while preserving the one-soliton dispersion. A contraction of the quantum W_3 algebra reproduces the canonical bracket and the semiclassical spectral spacing, but the inherited vacuum fails positivity. **(d)** The absolute phase requires physical soliton states and a boundary-compatible determinant or an equivalent dynamical calculation.

(c) For validation on the bad branch, vacuum elastic matching reproduces Toda's conditional one-body coefficient $1/(2\pi)$. The iterated Toda edge limit also gives a closed directional two-body coefficient. One continuum mixed difference agrees when evaluated using the Toda endpoint identity, but a ratio-dependent discrepancy survives in the full comparison. **(d)** Neither two-body equivalence nor uniform interchange with the fixed-momentum limit is established.

Audit status and evidence precedence. The README consolidation was `adb9116` (12:19 UTC on 4 November); the previous report was `446bf1a` (12:30 UTC), which set the good equation as the PI's priority. This edition covers the subsequent work through `bfc3944` (17:08 UTC). Where a package has a later independent audit, its corrections govern the results below. The Phase 2h, charge/action, determinant, frames/defects, Toda-dictionary and projected-measure audits remain in force. The table distinguishes independent audits from package checks.

Package (under `phase2/`)	Independent audit or scope of evidence
`toda_wh_and_kp_synthesis.tex`	`audit_phase2i.tex`
`frame_dictionary.tex`	`audit_frame_dictionary.tex`
`det_projected_oneloop.tex`	`audit_det_projected.tex`
`smx_candidates.tex`	`audit_smx.tex`
`defq_charges_fusion.tex`	`audit_defq.tex`
`onebody_dispersion_test.tex`, `onebody_toda_precision.tex`, `onebody_scheme.tex`	`audit_onebody.tex`, superseded where corrected by `audit_onebody_scheme.tex`
`good_renorm.tex`	Package checks
`good_twobody.tex`	`audit_good_twobody.tex`
`resonant_channels.tex`	`audit_resonant_channels.tex`
`defb_boundary.tex`	`audit_defb_boundary.tex`
`defect_bootstrap.tex`	Package checks
`twobody_toda.tex`, `twobody_cont.tex`	`audit_twobody_toda.tex`, `audit_twobody_cont.tex`
`uniform_limits.tex`	`audit_uniform_limits.tex`
`shiftcharge_oneloop.tex`, `shiftcharge_quantum_bridge.tex`	Package checks and internal cross-review in the bridge

Labels denote literature proof (a), literature heuristic (b), project derivation or computation (c), and unresolved interpretation or research judgment (d). The appendix records the verification scope and failures.

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1 Setting

The good equation is the PI's primary scattering problem. The classical derivation below starts with the bad branch used for Toda validation and gives the good-branch dictionary explicitly. **(c)** On the bad branch $u_b = 2(\log\tau)_{xx}$ and $k = \sinh\beta$. A soliton has velocity $v = \epsilon\cosh\beta$, with $\epsilon = +1$ for right-movers and $\epsilon = -1$ for left-movers. The potential is $u_b = \phi_x$ and the Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}\phi_t^2 - \tfrac{1}{2}\phi_x^2 + \tfrac{1}{2}\phi_{xx}^2 - \phi_x^3,$$

with $\pi = \phi_t$ and canonical symplectic form $\Omega = \int \delta\pi \wedge \delta\phi$.

On one soliton the charges are:

- mass $M = \int u_b\, dx = 2k$;
- ϕ-shift charge $\Pi = \int \pi = -vM$, conserved because $\phi \to \phi + c$ is a symmetry;
- momentum $P = -\int \pi\phi_x = \frac{2}{3}\epsilon k^3\cosh\beta$;
- energy $E = \frac{2}{15}k^3(5 + 4k^2)$.

These satisfy $dE/dP = v$ **(c)**. The corresponding good branch uses $k = \sin\alpha$, $v = \epsilon\cos\alpha$, cubic term $+\phi_x^3$ and dispersive term $-\phi_{xx}^2/2$ in the Lagrangian, and $E_g = 2k^3(5 - 4k^2)/15$. Checks: `errata_checks.py`, groups `energies` and `source_dictionaries`. For the BC field, $\varphi_{BC} = -\phi_b$; for CZ, $u_{CZ} = -3\phi_g$ and $\mathcal{L}_{CZ} = 9\mathcal{L}_g$. The corrected bad stable-band normal-ordering shift is $-3\hbar_B\sinh\beta/\pi$, not the historical positive sign. The old zero-scheme tables do not change (`errata.pdf`, M01–M03).

The two-soliton constants are

$$A_{\text{ov}} = \frac{1 + 3\coth^2\frac{\beta_1+\beta_2}{2}}{1 + 3\coth^2\frac{\beta_1-\beta_2}{2}}, \qquad A_{\text{ho}} = \frac{1 + 3\tanh^2\frac{\beta_1-\beta_2}{2}}{1 + 3\tanh^2\frac{\beta_1+\beta_2}{2}},$$

both in $(0, 1)$, and the position shifts are $\Delta x_1 = -k_1^{-1}\log A$, $\Delta x_2 = +k_2^{-1}\log A$ (soliton 1 starts on the left). **(c)** (Phase 1)

What a soliton S-matrix would mean here. **(c)** Neither chosen equation has the assumed relativistic boost symmetry; its long-wave vacuum dispersion is gapless. The bad vacuum also has unstable short-wave modes. The natural object is therefore the non-relativistic two-body S-matrix $S(P_1, P_2)$ of soliton asymptotic states, in the sense of Faddeev–Korepin [7]. **(c)** Regular classical two-soliton scattering is transmissive. **(d)** A unitary diagonal quantum soliton sector would therefore be a natural target, but is not constructed by that classical fact.

(c) For scalar diagonal scattering the Yang–Baxter equation is automatic. The non-trivial requirements are:

- unitarity;
- the correct semiclassical limit;
- analyticity and a crossing substitute;
- a bound-state (breather) spectrum consistent with poles.

2 Route A: semiclassical quantisation

2.1 The symplectic structure of soliton scattering

Proposition 2.1 (**(c)**). *Consider two well-separated solitons, with ϕ normalised by $\phi(-\infty) = c$. Up to exponentially small terms,*

$$\Omega = d\Pi_{\mathrm{tot}} \wedge dc + \sum_j dP_j \wedge dX_j + \sum_{i \neq j} \big[X_i > X_j\big]\, d\Pi_i \wedge dM_j.$$

The argument is as follows. π_i is localised at X_i, and there ϕ_j equals 0 or M_j according to whether soliton i is to the left or right of soliton j. The self-term $dP \wedge dX$ was derived in Phase 1. In sine-Gordon the analogous cross term vanishes, because the kink height $2\pi/\beta$ is fixed. Here $M = 2k$ varies with the soliton parameter.

Proposition 2.2 (**(c)**, `scripts/symplectic_scattering.py`). *The classical scattering map $X_j \mapsto X_j + \Delta x_j$ (with k_j and c fixed) preserves Ω, i.e.*

$$\sum_j dP_j \wedge d\Delta x_j = (\textit{cross term})_{\mathrm{in}} - (\textit{cross term})_{\mathrm{out}}.$$

This was verified to 40 digits for overtaking and head-on collisions at five parameter points each. It was also verified for the good *equation in its regular region (`scripts/consistency_checks.py`). Without the cross terms the identity fails by $O(1)$ amounts; for example, the left side equals 19.79 at $\beta = (0.9, 0.3)$.*

This is a non-trivial confirmation of the classical picture. It also shows that a semiclassical S-matrix built from the naive rule "$\Delta x_j = -\partial\delta/\partial P_j$" does not exist: the shifts are not curl-free in the canonical momenta.

2.2 The semiclassical phase

In each ordering region we choose Darboux coordinates by the rule

$$X_j' = X_j + \gamma_j \sum_{i \text{ left of } j} M_i, \qquad \gamma_j = \frac{d\Pi_j/dk_j}{dP_j/dk_j}.$$

The mirrored rule (normalising ϕ at $+\infty$) gives the same in–out differences. The shifts $\Delta x_j'$ are then curl-free. Their potential is the semiclassical phase δ, with $\Delta x_j' = -\partial\delta/\partial P_j$ and $S \simeq \exp(i\delta/\hbar)$, by stationary phase on two-particle wave packets [7].

Proposition 2.3 (**(c)**, checked semiclassical phase).

1. $\dfrac{dP}{d\beta} = \frac{2}{3}\,\epsilon k \sinh 3\beta$ *for the bad equation. For the good equation,* $\dfrac{dP}{d\alpha} = \frac{2}{3}\,\epsilon k \sin 3\alpha$ *with $k = \sin\alpha$; this vanishes at the cusp $\alpha = \pi/3$ found in Phase 1.*

2. *For both overtaking and head-on collisions, and both equations,*

$$\boxed{\delta(1,2) = \tfrac{2}{9}\,(\ell_1 - \ell_2)\,\log A_{12} + \tfrac{2}{3}\,M_1 M_2\,(v_1 - v_2)} \tag{1}$$

with $\ell = \epsilon\cosh 3\beta$ (bad) or $\ell = -\epsilon\cos 3\alpha$ (good). The remainder was first found by symbolic integration, then identified as $\frac{2}{3}M_1M_2(v_1 - v_2) = -\frac{2}{3}(\Pi_1 M_2 - \Pi_2 M_1)$ by exact matching.

3. Unitarity: $\delta(1,2) = -\delta(2,1)$.

4. Crossing-type relation: $\delta_{\rm ov}(\beta_1, \beta_2 + i\pi) = -\delta_{\rm ho}(\beta_1, \beta_2)$. *The shift $\beta \to \beta + i\pi$ maps a right-mover to a left-mover with $M \to -M$.*

Checks: `scripts/semiclassical_phase.py` and `scripts/consistency_checks.py`.

Remark 2.4.
- ℓ is proportional to the spectral parameter of the third-order Lax operator. In the uniformisation of [4], $\lambda = \frac{1}{2}(k^3 + k^{-3})$, with k^3 playing the role of $e^{3\beta}$.
- The dynamical part of δ is (difference of spectral parameters)$\times \log A$.
- The second part is a "mutual statistics" phase: an Aharonov–Bohm-like phase $\propto \Pi_1 M_2 - \Pi_2 M_1$ between objects carrying ϕ-shift charge and ϕ-winding. **(d)** (interpretation)

Remark 2.5 (KdV limit, **(c)**). For small amplitudes the topological term is of the *same order* as the KdV-type term $2\beta_1 \log A$ in $\partial\delta/\partial\beta_1$. At $\beta_1/\beta_2 = 1.7$, the full derivative divided by the KdV term tends to $0.564246498175\ldots$; the cross-only ratio is $-0.435753501825\ldots$ (`errata_checks.py`, `minor`). The literature heuristic **(b)** of Bonnemain and Doyon [2] identifies the quantum phase derivative with $P'_{\rm GHD}\Delta$, with $P_{\rm GHD} = \eta^2/2$. That identification reproduces our dynamical term but *misses the topological term.* The classical generalised hydrodynamics, which uses only Δx, is unaffected. **(c)**

2.3 Bohr–Sommerfeld: breathers as spectral-parameter strings

Bogdan and Charkina propose the semiclassical quantisation **(b)** of the Tajiri–Murakami breather of the bad equation by Bohr–Sommerfeld, $I_{\rm br} = N\hbar$, with an independently computed adiabatic invariant $I_{\rm br}$ **(a)** for the classical action, **(b)** for the large-quantum-number prescription [1, (64)–(65)].

Proposition 2.6 (**(c)**). *Set $\beta_{1,2} = a \pm ib$, which is Bogdan–Charkina's parametrisation $\kappa \pm ik = \sinh(a \pm ib)$. Then:*

1. *A_{12} equals their a_{12} [1, (33)]. In the stated localized window $0 < b < \pi/2$, regularity $A > 1$ holds precisely for $b > \pi/3$. The endpoint $b = \pi/2$ needs a separate localization condition (`audit_smx.pdf`).*
2. *$i \cdot \frac{2}{9}(\ell_1 - \ell_2) = I_{\rm br}$ exactly, with $I_{\rm br} = -\frac{4}{9}\sinh 3a \sin 3b$, in agreement with [1, (64)].*

Hence the coefficient of $\log A$ in (1) gives the classical breather action. Conditional on the semiclassical Bohr–Sommerfeld prescription,

$$\ell_1 - \ell_2 = -\tfrac{9i}{2}\, N\hbar, \qquad N \in \mathbb{N}, \tag{2}$$

which is a string condition in the spectral parameter ℓ.

This is the strongest cross-check in this report. It ties our two-soliton phase, built from Hirota shifts and the canonical symplectic form, to an independent quasiclassical calculation in the literature. Item 2 was checked at three parameter points to 10^{-16}.

Equation (2) is the form in which bound states appear in Bethe-ansatz solvable models. It suggests that ℓ is the variable in which an exact S-matrix has poles: $\ell_1 - \ell_2 \in -\frac{9i}{2}\hbar\mathbb{N}$, possibly with $\hbar$ renormalised. **(d)**

2.4 One-loop and candidate updates through November 2026

Continuum and projected determinant (Phases 2e, 2f, 2h). **(c)** Reflectionless tau-function phonon modes, dilogarithmic continuum integrals and direct potential-field spectral-density checks survive. The regulated co-moving trace requires its equal-mode-count cutoff endpoint and finite tadpole subtraction. The good-branch conditional fixed-momentum coefficient is $-2/\pi$ in that prescription. The new projected package proves

$$F|_P = e - vp = e_{\rm can} - vp_{\rm can} = -L_1, \qquad L_1 = vp - e,$$

where $E(P) = E_0(P) + \hbar_B F(P) + \cdots$. At fixed L_1, invariance between these two charge pairs is algebraic and unconditional; identifying L_1 with the complete effective action is not. Checks: `audit_det_projected.py`, checks 1–5, with its independent replacements for four tautological check groups.

(c) Translation, velocity and shift tangents have rank two. An auxiliary momentum direction gives a finite instantaneous projector and symplectic measure, but cannot remove the intrinsic kinetic-norm divergence caused by the velocity-dependent potential jump. The fixed-endpoint fluctuation domain, constraint evolution, Schur complement and time-boundary determinant remain missing **(d)**. Thus neither the conditional good coefficient nor its bad analytic continuation is a completed physical determinant. The vacuum matching that now reconciles the leading continued one-body coefficient is in Route D.

(c) At fixed physical momentum the original mode-charge conversion has logarithmic small-amplitude terms; the fixed-amplitude phonon limit alone does not establish a quantum Toda match (`matching.pdf`). The mode/canonical charge choice cancels from $F|_P$ but can still affect two-body frame terms. **(d)** The absolute exchange/Maslov phase and the complete physical scattering amplitude remain unresolved.

Bad branch and candidate tests. **(c)** The real stable-band trace in `oneloop.pdf` differs from the formal continuation used in the one-body comparison. Its extra resonance is a co-moving group-velocity condition; discarding unstable modes is not a construction of a full quantum field theory **(d)**. The elastic matching term in Route D does not repair that stable-band trace.

(c) The Gamma algebra and candidate poles at the recorded bad-equation Bohr–Sommerfeld levels survive `audit_smx.pdf`; no $N = 0$ pole occurs. **(d)** These are not an exact quantum spectrum or a derived S-matrix. The zero-background two-phase good Hirota breather class is singular **(c)**; this does not exclude every possible quantum bound state or other background **(d)**. The printed good-breather threshold, extra resonance line and smoothness claim are refuted: the genuine resonance limits are continuous but generally not differentiably smooth (`audit_smx.py`).

Ring equations and the residual unknown. **(c)** The corrected Darboux wall signs, conditional directional ring derivative and total charge sum rule are checked by `audit_smx.py`. The printed total Bethe derivative omits a self-winding term. **(d)** The quantum seam holonomy is not derived from wavefunction transport, so the proposed twisted equations remain conditional. Under compatible local analytic branches, a nonzero amplitude ratio and an assumed loop expansion, one real antisymmetric overtaking *function* parametrises the residual $O(\hbar^0)$ two-body unknown; crossing then determines its continued head-on channel **(c)**. This is an unresolved physical calculation **(d)**, not a licence to choose an arbitrary physical frame. The relation to the Toda constraints is in Route D. The audit refutes the printed J crossing offset and Stirling cancellation: direct quadrature gives crossing sum zero, not $\pi/2$. It also refutes the printed small-amplitude interface. None is used here to fix a physical constant.

2.5 Good-equation scattering at fixed momentum

What the vacuum condition fixes. **(c)** Write the finite elastic term as $\hbar\lambda \int u^2\,dx$. In the equal-mode-count trace, with the free-vacuum tadpole subtraction held fixed, the one-soliton slope C_g and static sound shift s_1 obey

$$C_g = \lambda - 2/\pi, \qquad s_1 = \lambda - 9/(4\pi), \qquad C_g - s_1 = 1/(4\pi).$$

Preserving the reference vacuum's classical elastic modulus would set $s_1 = 0$ and $\lambda = 9/(4\pi)$. The bare sequential on-shell phonon limit instead gives $-3/(2\pi)$ before finite subtraction. Inverse continuation of the Toda counterterm gives $+3/(2\pi)$ for λ and fails this static condition.

Checks in `good_renorm_trace.py`, `good_renorm_vacuum.py` and `good_renorm_lattice.py` distinguish the two orders of limits. **(d)** The static condition is a proposal. The exact spectral sound slope and canonical quantum lattice elasticity remain open.

Elastic response and clustering. **(c)** Write δ_0 for the classical phase (1). Differentiating the deformed scattering action at fixed P_1, P_2 gives the response per unit λ,

$$K_\lambda = 3\delta_0 + 4\sum_{j=1}^{2} P_j \Delta X_j'.$$

It is antisymmetric in the Darboux frame that reverses the chamber ordering on exchange. Retaining the symmetric boundary term $(\Pi_1 M_2 + \Pi_2 M_1)/2$ in a fixed chamber gives a nonzero symmetric response. The audit verifies the connected u^2 insertion and all 81 bilinear coefficients for transmission of a phonon past three solitons. Even the Darboux clustering test permits $F_{12} = c(P_1 M_2 - P_2 M_1)$. This example has rank two in its one-body functions and one antisymmetric coefficient. The 270-row grid leaves the absolute physical phase entries unset (`audit_good_twobody.py`). **(d)** A classification of the continuum ambiguity and the time-dependent, boundary-compatible two-body determinant are still missing.

Which solitons can appear as scattering states? **(c)** In the exact Y solution the local hierarchy charges are differences of functions of KP chord endpoints. They telescope when two chords concatenate. A Darboux generating function establishes this for the formal classical hierarchy, with independent Riccati checks through H_8. Light solitons ($0 < \alpha < \pi/3$) are kinematically protected against splitting into lighter solitons. Each interior heavy soliton ($\pi/3 < \alpha < \pi/2$) has an exactly on-shell pair of oppositely moving light solitons with the same charges. Varying the exact Y solution gives a localized growing perturbation with rate $\sqrt{3}(1/4 - v^2) > 0$. Thus already the classical heavy soliton is unstable. `audit_resonant_channels.py` checks the mode equation, kinematics and collision charges. **(d)** Its real one-loop mismatch does not supply a quantum decay width. Closure of a light-particle sector, including possible phonon emission, remains to be shown.

Does the shift charge fix the two-body coefficient? **(c)** Under the endpoint and wavepacket assumptions in `shiftcharge_oneloop.pdf`, continuity gives $\Pi = -M\dot{X}_{\rm cm}$. The conditional mean-profile mode insertion reproduces the canonical momentum derivative of the trace. At fixed P the kink height M also changes. Its contribution cancels the leading rational term inferred by varying Π alone. The separately computed Darboux recoil gives the Toda rational term in Route D and has different momentum dependence from F_{12}. Checks are `shiftcharge_modes.py`, `shiftcharge_algebra.py`, `shiftcharge_calibration.py` and `shiftcharge_projection.py`. **(d)** Neither $c_{\rm bad}$ nor $c_{\rm good}(\lambda)$ has been extracted.

3 Route B: quantum inverse scattering on the line

(d) *What would have to be true.* The Lax matrix $U(x, \lambda)$ would need an ultralocal r-matrix Poisson algebra, so that it can be regularised on a lattice and quantised à la Faddeev–Korepin–Sklyanin. Then a Bethe ansatz over a pseudo-vacuum with soliton excitations would give S.

Proposition 3.1 (**(c)**, `scripts/routes_BC_checks.py`)**.** *In canonical variables the Lax data of Phase 1 are* $q = \frac{3}{2}\phi_x - \frac{1}{4}$ *and* $r = \frac{1}{2}q_x + \frac{\sqrt{3}}{4}\pi$ *(good-equation real form; the bad one differs by* i*'s). Their brackets are*

$$\{q(x), r(y)\} = \tfrac{3\sqrt{3}}{8}\,\delta'(x-y), \qquad \{q, q\} = \{r, r\} = 0.$$

So the companion-matrix Lax connection is non-ultralocal.

Where it stalls. There are two known ways out, and neither yet gives a theory on the line.

1. Maillet's r/s formalism with lattice regularisation [14]. No such line-soliton construction has been established in this project.

2. The Miura (free-field) map. This is the route of Bazhanov–Hibberd–Khoroshkin [3] and Hikami [10]. It yields $U_q(\widehat{\mathfrak{sl}}_3)$ T- and Q-operators, but in the chiral conformal setting on a circle, with no particle sector.

Dimakis et al.'s Yang–Baxter maps [6] concern polarisations in the matrix case and are vacuous for the scalar equation. The "beyond Yang–Baxter" remark there is therefore not an obstruction here. For diagonal scattering, Yang–Baxter holds trivially. **(c)**

3.1 A canonical bracket and a relative scattering test

(c) The W_3 construction uses the second Gelfand–Dickey bracket, whereas the good potential theory uses the first. In `shiftcharge_quantum_bridge.pdf` a singular large-spin-background contraction produces the canonical bracket with an explicit $\hbar$ normalization. In the soliton variable ℓ the limiting Baxter arguments have spacing $\Delta\ell = -9i\hbar/2$, the unit in (2). The inherited highest-weight vacuum has a negative-norm mode under the required real-field involution (`shiftcharge_bridge_hikami.py`). **(d)** The normalized Baxter operators, positive representation and associated line-soliton eigenstates are unknown.

(c) An anchored classical current deformation with parameter g preserves isolated-soliton energy and momentum. To first order it changes the two-body action by $-g(P_1M_2 - P_2M_1)$ (`shiftcharge_bridge_deformation.py`). One-body dispersion and elasticity are therefore insensitive to this relative two-body direction. Gaussian sector tests retain an infrared logarithm even when finite-volume charge compensation costs vanishing energy (`shiftcharge_bridge_sector.py`). A conserved first-bracket density provides a regulator test. The naive three-site discretization fails it (`shiftcharge_bridge_hierarchy.py`). **(d)** The absolute phase of the undeformed theory and the interacting soliton state map remain open.

4 Route C: W_3, sl_3 affine Toda and the $U_q(\widehat{\mathfrak{sl}}_3)$ R-matrix

(d) *What would have to be true.* Boussinesq solitons would have to be identifiable as a limit of $a_2^{(1)}$ (complex) affine Toda solitons, whose S-matrices are the $U_q(\widehat{\mathfrak{sl}}_3)$ intertwiners of Hollowood [12] and Gandenberger [8].

Proposition 4.1 (**(c)**, `scripts/routes_BC_checks.py`)**.** *Take Hollowood's $a_2^{(1)}$ interaction constant [11, (1.19)], $A^{(12)}(\theta) = \frac{2\cosh\theta+1}{2(1+\cosh\theta)}$, between the two fundamental species. With $S = \beta_1+\beta_2$ and $D = \beta_1 - \beta_2$,*

$$A^{\mathrm{bad}}_{\mathrm{ov}}(\beta_1,\beta_2) = \frac{A^{(12)}(S)}{A^{(12)}(D)} \cdot \frac{\tanh^2(D/2)}{\tanh^2(S/2)}, \qquad A^{\mathrm{bad}}_{\mathrm{ho}}(\beta_1,\beta_2) = \frac{1}{A^{\mathrm{bad}}_{\mathrm{ov}}(\beta_1,\beta_2+i\pi)}.$$

Here $\tanh^2(\theta/2)$ is the sine-Gordon soliton–soliton constant. Equivalently,

$$A_{\mathrm{ov}} = \frac{G(S/2)}{G(D/2)}, \qquad G(y) = \frac{16\prod_{a=1,2}\sinh\frac{y+ia\pi/3}{2}\sinh\frac{y-ia\pi/3}{2}}{\sinh^2 y}.$$

So the Boussinesq interaction is built from exactly the $\mathfrak{sl}_3$ blocks, but in the form "$F(\beta_1 - \beta_2)/F(\beta_1+\beta_2)$". This is the structure of scattering off a particle *and its mirror image*. It suggests that a Boussinesq S-matrix might take the form

$$S(\beta_1,\beta_2) = \mathcal{S}(\beta_1-\beta_2)\,\mathcal{S}(\beta_1+\beta_2+i\pi)^{\pm 1}$$

with $\mathcal{S}$ built from $U_q(\widehat{\mathfrak{sl}}_3)$ and sine-Gordon factors. Such a form is reminiscent of boundary (reflection) amplitudes. That would connect to the "boundary affine Toda" structure in Bazhanov–Hibberd–Khoroshkin [3]. **(d)**

(c) The checked obstruction in the chosen rapidities is that the semiclassical phase (1) is not of this product form. Its weight $\frac{2}{9}(\ell_1 - \ell_2)$ is not a sum of functions of S and D, and its topological part has no Toda analogue.

Further obstructions:

- Toda solitons are relativistic with fixed masses m_a. In the Toda constant the mass term $4m^2 \sin^2(\pi a/3)$ appears where Boussinesq has K^4, so the species label is "replaced" by the wavenumber. No such limit or quantum dictionary has been constructed here; the displayed dispersion difference is not a universal impossibility proof **(d)** (`../paper/review_draft1.tex`, R12).

- Takács and Watts show that $a_2^{(1)}$ soliton–breather scattering is flawed both classically and quantum mechanically **(b)** [20].

(d) *Route C remains open*: the structural similarity is precise, but no dictionary is available. What is lost in the non-relativistic reduction is the difference form, and with it the standard crossing.

November candidate audit. **(c)** The classical block factorisation remains valid. The universal exclusion of mixed-argument quantum dilogarithms is refuted; a repaired residue argument excludes the specific nine KP-root ratios with constant coefficients. Exchange unitarity excludes a nonconstant smooth mirror factor, but permits a constant mirror factor and an exchange-unitary difference factor. The separate classical nonseparability test is not an exclusion of all quantum products. Checks: `audit_smx.py`, `families`. **(d)** No new affine-Toda quantum identification follows from these restricted exclusions.

5 Route D: Toda edge data, one-body matching and the two-body problem

Classical and finite-sea foundations (Phases 2b–2d). **(a)/(b)** The inherited quantum Toda and asymptotic Bethe framework is described in [9, 17, 19, 15, 18, 16]. **(c)** The implemented classical limit reproduces the Toda soliton phase with topological terms; its small-amplitude scaling recovers the bad-equation classical phase, with $\hbar_T = 3b\hbar_B$, $b = e^{-a/2}$ and $\kappa \sim \sqrt{3}\beta$. Checks: `toda_bethe_classical.py`, `toda_ba_vs_label.py`, `toda_scaling_fast.py`; the six dictionary normalisations are confirmed by the one-body scheme audit. This is a small-amplitude correspondence, not finite-amplitude quantum equivalence **(d)**.

(c) The Phase 2d head-on agreement and overtaking discrepancy from the Gamma candidate are results in the implemented classical frame. The claimed bound on successive raw phase changes, < 0.004, fails: the recorded changes are 0.006326, 0.009016, 0.004512 (`../scripts_inventory.md`). The extrapolated table checks pass separately. **(d)** Neither establishes a physical quantum S-matrix without the common continuum map.

One-body validation at the Toda edge. **(c)** Define F by $E(P) = E_0(P) + \hbar_B F(P) + \cdots$ and $C = \lim_{k\to 0} F(k)/|P_0(k)|$ within the specified edge/semiclassical extraction. This energy correction contributes at $O(\hbar^0)$ to the semiclassical phase. The initial `onebody_dispersion_test.pdf` found a factor-of-four mismatch. The higher-precision package now gives

$$C_T^{\text{num}} = 0.1591549414 \ \pm \ 8.24 \times 10^{-6}, \qquad C_T = \frac{3}{4\pi} - \frac{1}{4\pi} = \frac{1}{2\pi}.$$

The rounded-up envelope combines mesh/roundoff 5.68×10^{-8}, finite-$\hbar_T$ extrapolation 7.70×10^{-6}, and amplitude-tail 4.73×10^{-7} diagnostics. It is deterministic fit sensitivity, not a statistical confidence interval or rigorous error bound; the relative scale is about 5×10^{-5}. The unconstrained fit does not insert $1/(2\pi)$. The separate analytic calculation uses the exact Bethe derivative identity, momentum logarithm and harmonic vacuum normalisation. See `onebody_toda_precision.pdf` and `audit_onebody_scheme.pdf`. Checks are `onebody_prec_identities.py` and `onebody_prec_edge.py`. Uniform interchange with fixed nonzero momentum remains **(d)**, despite the stronger wording of the earlier one-body audit.

Vacuum elasticity resolves the leading mismatch. **(c)** In the inherited prescriptions, write $c_{\rm sound} = 1 + \hbar_B s_1 + \cdots$. The continued bad continuum and Toda give

$$C_{\rm ac} = \frac{2}{\pi}, \quad s_{1,\rm ac} = \frac{9}{4\pi}, \qquad C_T = \frac{1}{2\pi}, \quad s_{1,T} = \frac{3}{4\pi}.$$

Matching their vacuum elastic moduli, with no soliton target used, fixes the Hamiltonian density correction

$$\Delta\mathcal{H} = \hbar_B \lambda u^2, \qquad \lambda = s_{1,T} - s_{1,\rm ac} = -\frac{3}{2\pi}.$$

Since $P_0 = v \int u_0^2 dx$ and $|v| \to 1$, this shifts both C and s_1 by λ. Hence

$$C_{\rm matched} = \frac{2}{\pi} - \frac{3}{2\pi} = \frac{1}{2\pi}, \qquad C_{\rm ac} - s_{1,\rm ac} = C_T - s_{1,T} = -\frac{1}{4\pi}.$$

Checks: `onebody_scheme_continuum.py`, `audit_onebody_scheme.py`, and the read-only consolidation check `errata_checks.py`, `november_onebody`. Scheme invariance here means invariance under the common finite elastic subtraction, with the inherited origins/tadpole fixed. **(d)** The physical bad determinant, complete quantum equivalence, finite-amplitude matching and uniform limits are still open. The real stable-band resonance is not removed by this counterterm. For the good equation the conditional bare values are $C_g = -2/\pi$ and $s_{1,g} = -9/(4\pi)$, giving $C_g - s_{1,g} = +1/(4\pi)$ **(c)**; Toda forces no good-branch prescription **(d)**.

Frame dictionary: what is now fixed. **(c)** `audit_frame_dictionary.pdf` confirms the exact finite-sea endpoint, Jacobian and pressure-response identities, independently checks derivatives missing in the package, and confirms survival of the mixed classical-frame remainder in the edge limit. Its small-y error budgets require conditioning qualifications. One-body phase-reference shifts cannot cancel the mixed remainder. The exhibited antisymmetric addition evades only the incomplete one-body/reciprocity constraints; cluster separability and the microscopic Krein PDE exclude it as a physical Toda frame ambiguity. The parity/pole checks supply further restrictions (`audit_frame_dictionary.py`). The later `audit_twobody_toda.py` corrects the four-term characteristic source to $2f(u)B(v) - 2\sqrt{\pi}\, f(v)$, where B is the one-body edge response and $f = B'$ in the normalization of that audit. Homogeneous uniqueness with the same endpoint data still excludes the exhibited addition for this Toda observable.

(d) What remains is to identify the allowed one-body endpoint-response bilinears with continuum collective operators and the physical two-body phase. The continuum's single residual antisymmetric function in `audit_smx.pdf` is the same *kind* of two-body unknown, not an established equality to the rejected Toda example. Neither side licenses that example as a physical freedom on its own terms. One-body agreement does not determine the function.

Phase 2i: conditional asymptotics, not completed reconciliation. **(c)** The controlling `audit_phase2i.pdf` refutes the printed Fourier symbol, momentum transform/scale, inverse series, double transform, four-term reciprocity product, Darboux coefficient and kernel "if and

only if". With the missing normalisation restored, reversion and inverse-Laplace tests support the proposed edge coefficients. The formal integral identity

$$\frac{4}{\pi}\int_0^r \log(s/r)\frac{s^2}{1-s^2}\,ds = -\frac{4}{\pi}(\chi_2(r)-r), \qquad 0<r<1,$$

with $\chi_2(r) = [\mathrm{Li}_2(r)-\mathrm{Li}_2(-r)]/2$, and the proposed two-loop coefficient integral are confirmed as conditional calculations, not uniform Bethe asymptotics. Checks: `audit_phase2i.py`, `errata_checks.py:formal_integrals`. The corrected Krein convolution has independent numerical support. **(d)** Its finite-sea derivation still lacks zero-frequency matching and uniform endpoint control. The advertised $< 2\times10^{-4}$ data match uses one fixed-leading-coefficient estimator; alternative fits exceed it and can reverse subleading signs **(c)**. The later V2 tests below separate the logarithmic cancellation in the half-line calculation from the unresolved finite-sea limit.

5.1 Two-body validation and the order of limits

V1. Which Toda quantity is known? **(c)** Take the edge limit at fixed y_1, y_2, then let $y_1 = Y \to \infty$ at fixed $r = \sqrt{y_2/y_1} \in (0,1)$. The Phase 2i integral above gives the *forward* two-term residual. Subtracting the symmetric part $2(r+r^{-1})/\pi - \pi$, fixed by one-body endpoint responses, gives the antisymmetric directional coefficient

$$\vartheta_N^{(1)}(r) = \pi - \frac{4}{\pi}\chi_2(r) + \frac{2}{\pi}(r-r^{-1}).$$

Its value at $r = 1/2$ is $1.530527813279\ldots$. The directional head-on coefficient vanishes for this two-term observable, while a different antisymmetric endpoint representative retains a finite contribution. The large-Y and small-r fits are sensitive to the respective edge windows (`audit_twobody_toda.py`, `audit_twobody_cont.py`).

(c) In the lattice-matched bad continuum, the canonical recoil reproduces the rational term $2(r-r^{-1})/\pi$. The leading mixed rectangle $-(15\sqrt{2}-20)/\pi$ agrees when one uses the Toda endpoint-product identity to express it in terms of one-body data. The full conditional continuum candidate still differs from $\vartheta_N^{(1)}$ by a nonconstant function of r. The independent audit found 87 incorrect printed table cells and no calculation of the claimed two-body determinant (`audit_twobody_cont.py`). **(d)** V1 remains open pending a physical bad determinant and a common momentum and endpoint dictionary.

V2. Which asymptotics are uniform? **(c)** In the corrected half-line calculation the nonzero classical Taylor term $(\log Y)^2/Y$ cancels against the dressed-measure contribution. The endpoint moment vanishes. Under the matched-expansion assumptions the constant and $1/Y$ terms survive the endpoint layer. Independent inverse-Laplace convolution at $r = 1/2$ supports the formal $c_1(1/2) \simeq -0.021445$. Finite-sea fits still change the sign of the subleading coefficient across extrapolation bases. At fixed Y, the classical momentum tends to zero with h, whereas a fixed nonzero momentum sends Y to infinity. The phonon-first and soliton-edge limits of the half-line one-body slope give $3/(4\pi)$ and $1/(2\pi)$, respectively (`audit_uniform_limits.py`). **(d)** Interior direct-kernel bounds and these fits leave uniform finite-sea resolvent control and the interchange with fixed momentum unresolved.

Other discrete route. **(a)** Hikami's periodic quantum discrete Boussinesq construction [10] provides a separate algebraic starting point. **(d)** A line-soliton ground state, particle sector and thermodynamic scattering dictionary have not been constructed here. Section 3.1 gives the later canonical contraction and identifies the obstruction in the inherited vacuum.

6 Route E: defects, boundary charges and qualified fusion

(a) CZ construct type-I defects for the good equation [5]. **(c)** In project units $u_{CZ} = -3\phi_g$, $\mathcal{L}_{CZ} = 9\mathcal{L}_g$. The scattering root uses the unsigned vacuum parameter:

$$s = \sigma_{\mathrm{vac}}, \qquad q_D = -\frac{s}{2\sqrt{3}}, \qquad z = \left(\frac{v + s - \sqrt{3}k}{v + s + \sqrt{3}k}\right)^{\eta} = \left(\frac{p_b - q_D}{p_a - q_D}\right)^{\eta}.$$

The spectrum instead uses $c = \eta\sigma_{\mathrm{vac}}$. These must not be interchanged at $\eta = -1$. Checks: `errata_checks.py:roots_defect` and `audit_defq.py`. **(c)** The classical phase and regular transmission factorisation survive in this dictionary (`defect_semiclassics.pdf`); reflection and two-soliton decay were not covered.

New localized-charge and trace results. **(c)** `audit_defq.pdf` independently confirms the four classical localized charges and their conservation using the bulk equations and sewing conditions, and KP-edge factorisation of their vacuum jumps. The package's claimed numerical absorption test accidentally tests transmission because of an extra η; genuine absorption checks in both orientations pass in the audit. The quadratic boundary momentum trace is confirmed by independently solved sewing modes. The cutoff Casimir trace has a linear UV divergence requiring boundary subtraction; a bulk tadpole on the open half-lines cannot remove it. **(d)** The finite boundary prescription, complete mode/state normalisation and quantum Ward identities are not established. The restricted counterterm test is not an exhaustive classification of quantum boundary operators.

Spectrum and limits on the fusion obstructions. **(c)** The corrected determinant prefactor is sixteen times the historical one, and the admissible growing branch switches at $c = \sqrt{3}$. The remaining positive rate there is $\sqrt{3}/4$. For three distinct real KP roots the counts are generically $(0, 1, 2)$ and $(0, 1, 1)$ at the switch; no edge joins two vertices free of growing modes in this analysis. Thresholds and repeated roots require separate treatment. Checks: `audit_defq.py`, `errata_checks.py:defect_spectrum` and `november_defect`. **(d)** This is not a nonlinear stability/completeness theorem or an exclusion of a separately constructed resonance continuation.

(c) The pole used in the original argument against root shifts has zero residue. The errata addendum also corrects the energy-trace sign, the negative-wavenumber phase limit and the threshold rate.

Boundary subtraction and the triangle theorem. **(c)** The later `audit_defb_boundary.py` confirms the finite Casimir expressions in the specified subtraction and the stationary-root envelope identity. At $\lambda_g = 0$ the canonical energy holonomy is nonzero on *every* nondegenerate real KP triangle. The momentum holonomy has a single zero at the symmetric triangle. A finite bulk counterterm shifts the energy holonomy and can cancel the vector there at an exceptional coupling. Allowing the absorption rapidities to move cancels the counterterm contribution to the cycle balance. Consequently defect integrability does not select $\lambda_g = 0$. A classical transmission slope does not fix the finite quantum boundary counterterms. **(d)** The quantum absorption locus and a normalized boundary theory remain open.

What defect consistency says about F_{12}. **(c)** The proposed closed absorption/emission cycle contains a heavy chord on every nondegenerate real KP triangle. With only light external solitons and stable vacua there is no such cycle. Even on the formal triangle $F_{12} = c(P_1M_2 - P_2M_1)$ sums to zero, and a single-valued change of transmission phase preserves both the edge equations and two-defect composition. The additional rank of these constraints on c is zero (`defect_bootstrap_algebra.py`, `defect_bootstrap_sewing.py`, `defect_bootstrap_`

`transmission.py`). **(d)** Physical residues and a quantum boundary determinant have yet to be constructed.

KP bookkeeping (Phases 2g–2i). **(c)** Abstract chord closure, dipole differences and explicit orientation constants survive independent quadrature. Phase 2i explains the earlier orientation offsets algebraically, by a redefined kernel; it does not make them disappear from the raw integrals. Inversion antisymmetry suffices for exchange of the defined ratio bilinears, but the claimed necessity is false. **(d)** Branchwise analytic continuation is distinct from the real-part kernel and does not prove a global physical breather amplitude. Absolute phases, normalized defect states, quantum sewing and physical fusion remain open.

7 Consistency checks across candidates and orders

Claim	Audited status and check	Label
Classical phase	Symplectic/Darboux phase survives; corrected bad action and KdV ratio in `errata_checks.py`.	**(c)**
One-body Toda	$C_T = 1/(2\pi)$, analytic and numerical to about 5×10^{-5} relative; `onebody_prec_identities.py`, precision package.	**(c)**
Vacuum matching	$9/(4\pi)$ versus $3/(4\pi)$ fixes elastic counterterm; $C - s_1 = -1/(4\pi)$ agrees; `audit_onebody_scheme.py`.	Conditional **(c)**
Projected geometry	Finite instantaneous measure; intrinsic kinetic tails. $F\|_P = -L_1$ charge-pair invariant; trace/determinant identification conditional; `audit_det_projected.py`.	**(c)**/**(d)**
Toda frame	Endpoint/Jacobian identities and mixed remainder confirmed; arbitrary antisymmetric example excluded by cluster/Krein constraints; `audit_frame_dictionary.py`.	**(c)**
Continuum two-body	One residual antisymmetric function under local analytic and loop assumptions; common physical map not derived; `audit_smx.py`.	**(c)**/**(d)**
Phase 2i	Corrected-transform/formal-integral evidence survives; printed identities refuted, uniform asymptotics and strict fit error unproved; `audit_phase2i.py`.	**(c)**/**(d)**
Candidates/ring	Gamma algebra and bad candidate poles pass; restricted exclusions only; ring identities conditional, quantum holonomy unproved; `audit_smx.py`.	**(c)**/**(d)**
Defect charges	Classical charges, momentum trace, UV subtraction and growing counts pass; printed sign/phase/threshold/pole argument fail; `audit_defq.py`.	**(c)**
Good vacuum	Static subtraction would give $C_g = 1/(4\pi)$. Spectral matching is open. `good_renorm_trace.py`, `good_renorm_vacuum.py`.	**(c)**/**(d)**
Good two-body	Elastic response depends on chamber convention. Clustering retains the exhibited $P \wedge M$ term. `audit_good_twobody.py`.	**(c)**
Resonant channels	On-shell light daughters and a classical growing mode for each interior heavy soliton. `audit_resonant_channels.py`.	**(c)**
Boundary and fusion	All-triangle canonical energy result at $\lambda_g = 0$. No additional constraint on the exhibited coefficient. `audit_defb_boundary.py`, `defect_bootstrap_algebra.py`.	**(c)**
Shift charge	Height response cancels the proposed leading term. `shiftcharge_modes.py`, `shiftcharge_calibration.py`.	Conditional **(c)**

Claim	Audited status and check	Label
Quantum bridge	Canonical bracket and Baxter spacing, failed vacuum positivity, relative current deformation. `shiftcharge_bridge_hikami.py`, `shiftcharge_bridge_deformation.py`.	**(c)**
V1 and V2	Directional coefficient and conditional half-line expansion. Full continuum comparison and uniform finite-sea limit open. `audit_twobody_toda.py`, `audit_twobody_cont.py`, `audit_uniform_limits.py`.	**(c)**/**(d)**
Verification record	Phase 2d step bound and one-body audit tolerance still fail; tautologies replaced in audits, disabled Hirota check repaired; `../scripts_inventory.md`, errata addendum.	**(c)**
Full quantum theory	Physical bad determinant, exact S-matrix, full equivalence and uniform limits remain open.	**(d)**

8 Conclusions and ranked next steps

(c) Fixed-momentum elastic response, the heavy growing mode and the rank-zero defect test constrain what can be inferred about good-equation scattering. An explicit relative deformation explains how two Hamiltonians can have the same one-soliton dispersion while their scattering phases differ. **(d)** The main unsolved observable remains the absolute two-soliton phase on the light branch.

Good-equation priorities. The order below retains the PI's decision of 4 November, with each item reduced to the work still open.

1. **(d) Fix the physical vacuum condition.** Distinguish the exact spectral sound slope from the static elasticity proposal. A positive canonical representation and its finite subtractions would give an independent way to specify the good-equation coupling (Sections 2.5 and 3.1).

2. **(d) Determine the absolute light-soliton phase.** Construct physical states at fixed momentum and evaluate the boundary-compatible, time-dependent two-soliton determinant, or derive scattering from normalized operators in the same representation. The chamber response and relative deformation are checks on such a calculation. The seed coefficient is open.

3. **(d) Establish the quantum scattering channels.** Determine whether the light sector closes under scattering and how heavy resonances and phonons enter. The classical instability constrains which states can be used as external particles.

4. **(d) Complete the quantum defect theory.** Determine the finite boundary terms, state normalization, quantum absorption locus and residues. The fixed-edge triangle theorem and the stable-vacuum bootstrap delimit what the existing constraints can fix.

5. **(d) Select a candidate and derive ring quantisation.** Determine physical poles, branch data and the quantum seam holonomy in the same light-particle state space.

Supporting validation on the bad branch and in Toda.

V1. **(d)** Compare the full two-body phases using common physical momenta and endpoint references. The directional formula and mixed rectangle in Section 5.1 are benchmarks.

V2. **(d)** Establish uniform finite-sea resolvent and endpoint bounds that would justify the fixed-momentum limit. The half-line coefficients and refits do not provide this bound.

V3. **(d)** Derive a physical prescription for the bad-equation determinant, or obtain a PI choice before extending the continued trace.

Relation to the paper. Draft 3 incorporated the audited one-body, determinant, Toda, candidate and defect results. Its independent review required eleven corrections. Draft 4 applies all eleven, including the Krein source, symplectic signs, projector inputs, one-loop units and tadpole sign, and the lab-frame collision operator. It distinguishes absence of real-rate growing defect modes from full spectral stability and corrects the stated scope of the paper's checkers. The later V1 audits, defect bootstrap and shift-charge notes postdate Draft 4. See `../paper/review_draft3.pdf`, `../paper/boussinesq_soliton_scattering.pdf` and `../paper/scripts/paper_draft4_checks.py`.

A Phase 2 Write-ups and Verification Scripts

This revision changes the evidence status of the quantum synthesis; it does not introduce new external references or claim to have reread the inherited bibliography. Literature-reading scope is recorded in the independent notes. Write-ups and checkers are in `phase2/` and `phase2/scripts/`, with defect scripts also in `corrigan-zambon/`. Success markers apply only after actual assertions pass. The inventory and errata addendum describe the historical generators and failed checks. For this edition we ran the Python scripts added or modified between `446bf1a` and `bfc3944` (42 in Phase 2 and three in the paper), including the six independent audits on page 2. We also ran `errata_checks.py` and the Phase 1, defect-directory and paper scripts. Runs used separate copies of the sources and the scripts' default cache modes. The Phase 1 figure generator is recorded separately from the verification scripts. The list below also indexes earlier Phase 2 calculations whose results were carried forward from their audits.

- **Phase 2a** (`report.pdf`)**:** `scripts/symplectic_scattering.py`, `scripts/semiclassical_phase.py`, `scripts/fit_R.py`, `scripts/consistency_checks.py`, `scripts/routes_BC_checks.py`.
- **Phase 2b** (`toda_route.pdf`)**:** `scripts/toda_bethe_classical.py`, `scripts/toda_bethe_quantum.py`, `scripts/toda_ba_vs_label.py`, `scripts/toda_scaling_fast.py`, `scripts/toda_quantum_corrections.py`.
- **Phase 2c** (`smatrix_ansatz.pdf`, `defect_semiclassics.pdf`)**:** `scripts/smatrix_rejected_forms.py`, `scripts/smatrix_semiclassical_unitarity_crossing.py`, `scripts/smatrix_poles_bootstrap.py`, `scripts/defect_smatrix_good_continuation.py`, `corrigan-zambon/defect_*.py`.
- **Phase 2d** (`toda_edge.pdf`, `oneloop.pdf`)**:** `scripts/toda_edge_ba.py`, `scripts/toda_edge_twoterm.py`, `scripts/oneloop_step0_wellposedness.py`, `scripts/oneloop_phase_shifts.py`, `scripts/oneloop_soliton_energy.py`, `scripts/oneloop_breather.py`.
- **Phase 2e** (`oneloop_phase.pdf`)**:** `scripts/oneloop_good_phonon_scattering.py`, `scripts/oneloop_good_twosoliton_phase.py`, `scripts/oneloop_good_limits_cusp_resonance.py`.
- **Phase 2f** (`matching.pdf`)**:** `scripts/matching_conventions.py`, `scripts/matching_toda.py`, `scripts/matching_analysis.py`.
- **Phase 2g** (`oneloop_defect_phase.pdf`)**:** `scripts/oneloop_defect_scattering_and_stability.py`, `scripts/oneloop_defect_phase_and_fusion.py`.

- **Phase 2h (`charge_frame_master.pdf`):** `scripts/charge_and_frame_audit.py`, `scripts/kp_master_and_toda_edge.py`.
- **Independent audits:** `independent_phase2h_audit.pdf`, `audit_charges_action.pdf`, `audit_determinant_foundations.pdf`, and `audit_frames_defects.pdf`, with the corresponding `audit_*.py` scripts listed in the consistency table.
- **Follow-up audits:** `audit_projected_measure.pdf` and `audit_toda_dictionary.pdf`, checked by the corresponding named scripts and saved JSON results.

November evidence index. Earlier November checkers share the basenames of their audit notes on page 2. The historical `audit_onebody.py` still fails its original tolerance. Its corrected interpretation is in `audit_onebody_scheme.pdf`. The `onebody_prec_*` scripts check the numerical extraction and its analytic edge value. The `onebody_scheme_*` scripts check the vacuum elastic match. `errata_checks.py` checks the consolidated corrections.

November failures recorded. **(c)** The Phase 2d 0.004 step bound fails at three recorded steps; the earlier one-body audit's 3×10^{-6} tolerance fails with residual $2.2546734882 \times 10^{-4}$. Finer extrapolation and independent inversion support the limiting identity, but do not make the unchanged script pass. Four `det_*` tautology groups and one `frame_identities.py` tautology were replaced by independent checks in their audits. The disabled Hirota `orTrue` assertion was repaired in the hygiene package. The Phase 2i, S-matrix and defect refutations, including sign/branch errors and tests that passed without testing their printed claims, are itemised in `errata.pdf`. Failed audit-development checks remain in the individual audit notes.

Failures recorded in the subsequent work.

- **(c)** The good two-body counterterm response lost a symmetric boundary term when it was described independently of chamber. The resonant-channel audit corrected the printed energy slope, singular-example values and Gamma fusion factor. The generating-function proof replaces the invalid commuting-background argument for hierarchy additivity.
- **(c)** The boundary audit refuted the fixed-momentum transmission sign and the inference that integrability selects the bulk subtraction. The defect cycle adds no condition on the displayed antisymmetric coefficient.
- **(c)** The Toda frame audit's printed four-term source differentiates the wrong endpoint. The bad-continuum audit corrected the profile and stable-band signs, the subleading rectangle and the grid tables. The continuum audit flagged an incorrect $r = 1/2$ example in a concurrent draft of the Toda audit. The committed Toda audit gives the corrected value in Section 5.1.
- **(c)** Freezing the kink height fails the shift-charge calibration. The instantaneous projection still violates fixed endpoints at the tested collision. The inherited highest-weight vacuum fails positivity, and the three-site regulator fails the higher-charge bracket test.

The named package and audit notes retain their individual development failures and repairs.

Failures along the way (Phase 2a; later phases record their own failures in each write-up).

- The direct symbolic simplification of the symplectic identity timed out, so we switched to 40-digit numerics.
- The good-equation phase first used the wrong sign of ℓ, and the test failed until it was corrected.

- A tolerance set below double precision caused spurious failures.
- The KdV-limit comparison with Bonnemain–Doyon *does not* agree. This is recorded as a finding, not a passed test.

Failures recorded by the independent quantum audits.

- Initial PDF builds had overfull filename lines and a temporary text-group scoping error; line breaking and group closure were repaired before the final rendered-page inspection.
- During an earlier report revision, the Phase 2h audit's source-snapshot guard failed because the report was edited while the script was running. The guard was retained; verification was repeated after editing stopped.
- The classical momentum derivatives in this report omitted the orientation factor ϵ and therefore applied only to right-movers. Both signed identities are now checked in `audit_projected_measure.py`.
- Raw-phase antisymmetry, zero diagonal, and orientation-independent master identification failed independent continuum checks.
- The displayed defect determinant prefactor and the complete growing-mode pair failed checks away from zeros and between the old sampled parameters.
- The proposed Toda residual was used to define part of its purported derivation; saved extrapolation alternatives changed the subleading sign. The claimed edge-reference explanation of a ratio-dependent slope also failed.
- Potential amplitude modes have nonlocal tails, invalidating the stated localization-only cancellation argument. Equal endpoint frequencies do not by themselves establish localized determinant cancellation.
- Profile, weighted-sum, and derivative-sign errors in intermediate charge formulas were found although the final conditional charge derivative passed. Boundary counterterms and localized defect momentum were not accounted for in the claimed physical absorption argument.

Development failures, repairs, precision, and the scope of each successful `ALL CHECKS PASSED` run are recorded in the audit notes. A successful algebraic script does not certify the unresolved physical identifications.

Reference verification and reading scope. **(c)** Entries below are inherited sources, with metadata corrections taken from `errata.pdf` and `../phase0/reference_audit.md`. No external full text or abstract was newly read for this consolidation. The errata records relevant full-text passages read for CZ and BC, abstract/metadata-only verification for many other entries, and access limits; those distinctions are retained here. Literature labels describe the inherited source scope, not a fresh literature proof. In particular the BC action is classical (a), its quantisation prescription semiclassical (b), and exact quantum pole identification (d).

References

[1] M. M. Bogdan, O. V. Charkina, arXiv:2502.14695, *Low Temp. Phys.* 51 (2025) 763–773, doi:10.1063/10.0036799.

[2] T. Bonnemain, B. Doyon, *SciPost Phys.* 18 (2025) 075, arXiv:2402.08669.

[3] V. V. Bazhanov, A. N. Hibberd, S. M. Khoroshkin, *Nucl. Phys. B* 622 (2002) 475, arXiv:hep-th/0105177.

[4] C. Charlier, J. Lenells, arXiv:2204.02365.

[5] E. Corrigan, C. Zambon, *Adding integrable defects to the Boussinesq equation*, *J. Phys. A* 56 (2023) 385701, arXiv:2306.03772. doi:10.1088/1751-8121/aceec9.

[6] A. Dimakis, F. Müller-Hoissen, X.-M. Chen, *Phys. Scr.* 94 (2019) 035206, arXiv:1805.09711.

[7] L. D. Faddeev, V. E. Korepin, Quantum theory of solitons, *Phys. Rep.* 42 (1978) 1, doi:10.1016/0370-1573(78)90058-3.

[8] G. M. Gandenberger, arXiv:hep-th/9703158.

[9] M. C. Gutzwiller, *Ann. Phys.* 133 (1981) 304–331, doi:10.1016/0003-4916(81)90253-0.

[10] K. Hikami, *Nucl. Phys. B* 604 (2001) 580, arXiv:nlin/0102021.

[11] T. J. Hollowood, *Nucl. Phys. B* 384 (1992) 523, arXiv:hep-th/9110010.

[12] T. J. Hollowood, *IJMPA* 8 (1993) 947, arXiv:hep-th/9203076.

[13] P. R. Johnson, *Nucl. Phys. B* 496 (1997) 505, arXiv:hep-th/9611117.

[14] J.-M. Maillet, Kac–Moody algebra and extended Yang–Baxter relations in the O(N) non-linear sigma model, *Phys. Lett. B* 162 (1985) 137, doi:10.1016/0370-2693(85)91075-5. [not read]

[15] F. G. Mertens, *Z. Phys. B* 55 (1984) 353–360, doi:10.1007/BF01304087.

[16] M. Pustilnik, K. A. Matveev, *Phys. Rev. B* 92 (2015) 195146, arXiv:1507.05639.

[17] E. K. Sklyanin, *Lect. Notes Phys.* 226 (1985) 196–233, doi:10.1007/3-540-15213-X_80.

[18] R. Siddharthan, B. S. Shastry, *Phys. Rev. B* 55 (1997) 12196, arXiv:cond-mat/9611221.

[19] B. Sutherland, *Rocky Mountain J. Math.* 8 (1978) 413–430, doi:10.1216/RMJ-1978-8-1-413.

[20] G. Takács, G. M. T. Watts, *Nucl. Phys. B* 547 (1999) 538, arXiv:hep-th/9810006.